

Civic.ai - A Citizen Centric AI-Powered Government Service Assistant

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Declaration

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Abstract

Accessing government services in Bangladesh is often difficult due to confusing procedures, lack of clear and centralized information and reliance on middlemen. This prevents many citizens from accessing these services efficiently, resulting in unnecessary time delays and financial burdens. This thesis aims to solve this problem by developing an AI-Powered interactive system that provides citizen centric guidance for certain government services. which will act as a digital bridge between citizens and government services. This system will inform users about required procedures and credentials of the respective services by offering dialogue and question-answering sessions. The outcome is a user-friendly, accessible system that empowers citizens with correct information, reduces dependency on intermediaries and simplifies interactions with government offices.

Keywords: Artificial Intelligence (AI), Government Services, Intermediaries, AI-Powered interactive system, Citizen centric guidance, Digital bridge, Dialogue and question-answering sessions, User-friendly and accessible system.

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Chapter 1

Introduction

This chapter describes the background, core motivation and objectives of our study. It begins by contextualizing the existing challenges citizens face in accessing public services in Bangladesh, followed by the rationale for developing an AI powered solution. This is how it formulates the problem statement and defines the specific goals and scope of the project. It also reflects the significance of this work in the context of e-governance and public service transformation. Finally, the chapter briefly outlines the proposed methodology, research phases and expected outcomes that establish our groundwork for subsequent technical and analytical discussions.

1.1 Background

Digitalization in government services has become an essential part of the modern world. Countries are adopting modern technologies to make government services more easily accessible to its citizens, including Bangladesh. In Bangladesh, government agencies have launched various online portals intended to streamline public service delivery. Yet citizens frequently encounter scattered information, unclear procedures and lack of guidance while using these systems. In the meantime, progress in artificial intelligence (AI), especially after the development of Transformer architecture [2], have revolutionized the understanding of natural-language for computers. This revolutionary technology has become the backbone of modern Large Language Models (LLMs) while inventions like LLaMA [19] and LaMDA [14] proved that transformer based models can engage in interactive dialogue. Later, BanglaBERT [9] made these architectures also applicable to low-resource languages like Bangla.

These technologies have made AI-powered assistants capable of understanding citizen queries in both English and Bangla. It has also made it possible to retrieve accurate procedural information and interact conversationally. Experiments in AI-driven civic services [1], [12], [17], [21], [24], [25], [27], [30] have shown improved efficiency and citizen engagement in different countries. Yet, Bangladesh still doesn't have such a citizen-centric AI-powered platform. That's why, this study proposes Civic.ai, an AI-powered government-service assistant, which can provide reliable, contextual and inclusive information and guidance to all citizens.

1.2 Rational of the Study or Motivation

The fundamental motivation for this research originates from the imbalance between the availability of digital infrastructure and the citizen's ability to use it effectively. People often depend on unofficial intermediaries to complete processes, resulting in delays, financial burden and reduced trust in government systems.

From a technical viewpoint, the emergence of Generative AI [14], [19] and Retrieval-Augmented Generation (RAG) [8], [16], [23], [29] raise factual, up-to-date responses drawn from trusted sources. Integrating such technology into public service chatbots can shift static webpages into interactive guidance systems which are dynamic and lower cognitive barriers for non-technical users.

Academically, Civic.ai combines several state-of-the-art research areas such as transformer architectures [2], low-resource multilingual modeling [9], conversational AI [11], [15], [18], RAG frameworks [8], [16], [23], [29], and evaluation methodologies [3], [4], [5], [6], [7], [10], [13], [20], [22], [26], [28]. Such alignment makes the project both socially impactful and scientifically valuable. Socially it supports inclusive governance and scientifically it evaluates how bilingual generative models can deliver trustworthy public information in low-resource environments.

1.3 Problem Statement

Despite having existing E-government initiatives, citizens in Bangladesh continue to face persistent obstacles:

- **Dispersed Information:** Procedures and document lists often differ across government sites causing confusion and not having a central source of information that citizens can rely on.
- **Language Barriers:** Most instructions remain in English, limiting accessibility for Bangla speaking users.
- **Reliance on Intermediaries:** Many citizens depend on brokers to complete forms. Which increases cost and risk of fraud.
- **Lack of Interactivity:** Current portals offer static FAQs instead of dynamic and no such personalized guidance is available for curated needs.
- **Waste of time:** Citizens often need to go to different offices and desks and stand in queues to perform some simple task or just to gather information which consumes nearly a whole day.

These challenges return inefficiency, misinformation, and inequitable access. Existing digital systems fail to provide solid, bilingual dialogue that adapts to citizen needs.

Research Problem: How can an interactive AI powered assistance system with bilingual and retrieval-based features be built to give clear, accurate and easily comprehensible guidance based on context-aware information about government services in Bangladesh?

1.4 Objective

Overall Objectives

To design and evaluate Civic.ai, a citizen-centric AI assistant that utilizes open-source LLMs and RAG to bridge the information gap between citizens and government services.

Specific Objectives

- Explore transformer and attention-based language architectures [2], [9], [14], [19] to determine suitability for bilingual comprehension.
- Adapt or fine-tune open-source LLMs like LLaMA [19] and BanglaBERT [9] for English–Bangla interactions.
- Integrate RAG pipelines [8–11] to connect the model with curated government knowledge bases.
- Develop a chatbot interface capable of multi-turn conversation, intent recognition, and real-time information retrieval [11], [15], [18].
- Evaluate accuracy, coherence, and factual reliability using multiple benchmarks [3], [4], [5], [6], [7], [10], [13], [20], [22], [26], [28].
- Analyze ethical, accessibility, and governance implications of deploying generative AI in specific public services [1], [12], [17], [21], [24], [25], [27], [30].

Expected Impact

If Civic.ai is successfully implemented as planned, citizen dependency on middlemen will be significantly reduced and over time, such intermediaries may no longer be necessary. The system will provide citizens with valid and consistent information, allowing them to ask questions and receive guidance about government services anytime, from the comfort of their homes. This will not only save time but also serve as a blueprint for AI-powered civic services in other developing nations.

1.5 Methodology in Brief

For Civic.ai we intend to follow a five-phase approach which will integrate exploratory study, model development and evaluation within a unified framework. In phase one, we started with a comprehensive study and review of existing works on transformers, LLMs, chatbots, RAG, evaluation metrics and AI governance. In phase two, we will do a ‘need analysis’ by conducting interviews and surveys with

citizens and government officials to understand the major gaps in more detailed and quantitative form. In phase three, based on these insights, we will fine tune a bilingual base model combining LLaMA [19] for English and BanglaBERT [9] for Bangla with government-specific datasets and enhanced using a hybrid dense-sparse RAG module [8], [16], [23], [29] to ensure factual solidity and contextual accuracy. This integrated system will operate through a streamlined pipeline following user query, language detection, embedding, retrieval, generation and response delivery while supporting code-mixed English-Bangla interactions. In phase four, model evaluation will employ benchmarks such as GLUE [3], SuperGLUE [5], XQuAD [4], TyDi QA [7], BERTScore [6], HELM [13], MT-Bench [20], and RAGAs [22], supported by small-scale user testing to evaluate the overall clarity and satisfaction of the responses. Finally, in phase five, we intend to analyze the results against baselines and ethical considerations like privacy and bias [1], [12] will also be addressed in this process. We also aim to formulate technical plus policy recommendations to guide future deployment of AI-powered citizen-service systems in Bangladesh.

1.6 Scopes and Challenges

Scopes

The project focuses on developing an AI-driven information guidance system designed to assist Bangladeshi citizens in accessing essential government services such as passport applications, national ID (NID) services, land tax and other government services. The system emphasizes bilingual interaction (Bangla and English) through a text-based chatbot interface, with future potential for multimodal integration, including voice and document-based input. Beyond information delivery, the long-term scope of the project envisions enabling the system to perform automated verification processes that currently require manual handling by government offices. This would include verifying user credentials, cross-checking submitted information with public databases, identification and verification of documents and guiding users through end-to-end service completion. Such automation aims to reduce human dependency, minimize processing delays, and enhance transparency and accessibility in citizen-government interactions.

Challenges

Key challenges include several critical factors. Finding the right data is tough because it's often locked away in unstructured documents scattered across various departments [25], [30]. Language complexity poses another barrier, as Bangla's rich morphology and the scarcity of high-quality pre-training resources compromise model performance [4]. Issues of factuality and hallucination arise since large language models (LLMs) may produce incorrect or fabricated information [8], [15], [16], [23], [29]. Ethical and governance concerns also emerge, especially regarding data privacy and public trust in AI-driven systems [1], [12]. Finally, infrastructure limitations including the need for substantial computational resources for both training and deployment present additional challenges [8], [19].

Chapter 2

Literature Review

In this chapter, we are presenting the reviews of the basic research and recent progress in transformers, large language models (LLMs), chatbots, retrieval-augmented generation (RAG) and AI-driven governance systems. It explores the evolution of architectures, bilingual NLP techniques as well as factual retrieval methods that shape the proposed Civic.ai framework. This discussion brings together the findings from the studies, surveys and real world tests to show how earlier research has helped improve transformer based chat systems, support multiple languages and evaluate the technologies. We concluded this chapter by summing up the main points and pointing out what is still missing, the need for a system that can provide useful information and focus on helping the citizens so that they can get hasslefree government services in the country.

2.1 Preliminaries

The modern revolution in language-based artificial intelligence originates from the introduction of the Transformer architecture by Vaswani et al. [2]. This model replaced sequential recurrent processing with a self-attention mechanism which enables parallelization and the capture of long dependencies within text. The innovation allowed models to scale to billions of parameters, forming the core of later large language models (LLMs).

Subsequent generative models later expanded this foundation. LLaMA [19] demonstrated that open and efficient foundation models can achieve competitive performance using smaller data and computational budgets, while LaMDA [14] from Google works on fine-tuning transformer models specifically for dialogue or conversational applications, emphasizing two major challenges: safety and factual grounding.

For low resource languages such as Bangla, BanglaBERT [9] provided a great step by pretraining on large-scale monolingual corpora. These models establish the bilingual foundation upon which Civic.ai can operate.

In conversational AI, end to end task oriented dialogue models [15] have evolved to handle multiple tasks within a unified framework, using modern day transformer based encoders and decoders to generate coherent responses. Chain-of-Verification (CoVe) [11] proposes a reasoning strategy that reduces hallucinations by cross-

verifying outputs across multiple generated chains. Research that is specific to Bangla conversational systems [8] shows that transformers can effectively power domain-specific chatbots, providing valuable guidance for our bilingual context.

The next theoretical cornerstone is Retrieval-Augmented Generation (RAG) [23], which combines a retriever module (to fetch relevant documents) and a generator (to compose the responses). Later surveys [16], [25], [29] expanded on this concept, which represents best practices, architectural improvements and applications for multilingual and fact based question-answering systems. These studies collectively suggest that hybrid retrieval plus generation pipelines offer both scalability and factual robustness which makes them suitable for real-time public service dialogue systems like Civic.ai.

2.2 Review of Existing Research

2.2.1 Advances in Transformer and LLM Architectures

Vaswani et al. [2] introduced the Transformer, which is an encoder-decoder model that replaced recurrent and convolutional layers with new multi-head self-attention mechanisms and that enables faster training and much faster performance in machine translation tasks. This introduced positional encoding, scaled dot product attention and feed-forwarded layers trained on the publicly available WMT 2014 English-German and English-French datasets. The Transformer gained BLEU scores of 28.4 and 41.0 respectively. It used 8 GPUs and took 3.5 days to train and that makes it cost effective as well.

LLaMA [19] introduced open-source scaling laws and optimized performance by training on massive quantities of data (up to 1.4T tokens). This architectural focus on efficiency and publicly available data makes it ideal for governmental deployment where transparency, privacy and cost are crucial.

LaMDA [14] enlarges the transformer paradigm by integrating safety, dialogue-optimized pretraining and sensibility metrics to produce conversational coherence. The family includes models up to 137 billion parameters and pretrains on almost 1.56 trillion words.

For low-resource languages like Bangla, BanglaBERT [9] presents a significant development in natural language processing. They developed the first Bangla-only pretrained language model, BanglaBERT, which was pretrained on over 2.18 billion Bangla tokens retrieved from the large-scale Bangla2B+ corpus, curated from 110 diverse websites and heavily preprocessed. Using the ELECTRA architecture with Replaced Token Detection (RTD) as its pretraining objective, BanglaBERT achieved superior performance across multiple tasks including sentiment classification, natural language inference, named entity recognition and question answering. This demonstrates that dedicated monolingual pretraining, even in a low-resource context can lead to more efficient and effective models. The associated BLUB benchmark further provides a standardized evaluation framework, establishing a foundation for future research in Bangla NLP.

Bangla-specific innovations [9] bridge the low-resource gap by improving tokenization and contextual embeddings for morphologically rich languages. It enhances representation quality and performance across Bangla NLP tasks through transformer-based pretraining tailored specifically for the language.

2.2.2 Conversational AI and Chatbot Development

Recent surveys on End-to-end Task-oriented Dialogue (EToD) systems [18] propose a taxonomy that distinguishes between Modular and Fully EToD paradigms. Empirical evidence from this survey highlights that Modular EToD architectures built on pre-trained language models deliver the best results across multiple benchmark datasets. For instance, the SPACE3 model achieved a Combined score of 111.0 on the MultiWOZ 2.0 dataset, the highest reported among PLM-based systems. On CamRest676 (a small-scale dataset with 408 training, 136 validation, and 136 test dialogs), SPACE3 attained a 97.7 Match score, while BORT reached the highest Success (89.7) along with BLEU (25.9) scores. In the Fully EToD category, the MAKER model recorded a BLEU of 25.9 and Entity F1 of 71.3% on the Stanford Multi-Domain (SMD) dataset [6]. Such metrics demonstrate that transformer-based dialogue systems achieve both high coherence and task completion rates across domains.

The Chain-of-Verification (CoVe) method [6] introduces a meta-reasoning approach that significantly reduces hallucination in large language models. On the Wikidata test set of 56 list-based questions, CoVe (two-step) doubled precision from 0.17 to 0.36 and lowered the average number of hallucinated answers from 2.95 to 0.68. For closed-book QA on 418 MultiSpanQA questions, the factored CoVe variant boosted F1 by 23% (0.39 $\rightarrow$ 0.48). In long-form biographical generation, the “factor + revise” setup raised the FACTSCORE from 55.9 to 71.4, a 28% gain over the few-shot baseline and notably outperforming both ChatGPT (58.7) and PerplexityAI (61.6). These results quantify CoVe’s effectiveness in enhancing factual accuracy through self-verification.

Localized research also supports the potential for Bangla chatbots. The Transformer-Based Bengali Chatbot study [11] demonstrated a major performance jump over traditional Seq2Seq approaches on a 2,000-sample Bangla QA dataset. The Transformer model that had a BLEU score of 85.0 that actually outperformed the Seq2Seq baseline (23.5), LSTM (17.1), and Simple RNN (12.3) models. Training for 120 epochs with a batch size of 28 returned a final loss of 0.002, indicating exceptional fit and stability. These metrics validate transformers as a robust foundation for bilingual government-service assistants like Civic.ai, which must handle English-Bangla code-switching and intent variation with high precision.

Overall, the evidence from [11], [15], [18] confirms that modern transformer-based dialogue models not only achieve state-of-the-art task success rates but also benefit from explicit verification mechanisms to ensure factual integrity which is a critical requirement for public information systems such as Civic.ai.

Recent surveys [18] reveal that modern chatbots integrate natural language un-

derstanding (NLU), dialogue management and natural language generation (NLG) within end-to-end frameworks that often utilize transformers. Chain-of-Verification [15] introduces a meta-reasoning layer that validates factual consistency, an essential concept for public information assistants.

Localized work such as the Transformer-Based Bengali Chatbot [11] highlights challenges like code-switching and low-resource data. These insights directly inform Civic.ai, which must maintain robustness across mixed English–Bangla utterances.

Globally, conversational AI research increasingly emphasizes reducing hallucinations, improving grounding and tailoring responses to user context, issues mirrored in governance chatbots [1], [17]. Integrating dialogue systems with RAG can substantially improve factual correctness while maintaining conversational fluidity [8], [16], [23], [29].

2.2.3 Retrieval-Augmented Generation (RAG) and Knowledge Integration

The foundational RAG framework [8] introduced a hybrid model combining parametric memory (BART-Large generator, 400 M parameters) and non-parametric memory (a dense index of 21 million Wikipedia chunks - 15.3 billion values). Despite being far smaller than the T5-11B closed-book model, RAG achieved state-of-the-art accuracy on multiple open-domain QA benchmarks: 44.5 EM on Natural Questions, 45.2 on WebQuestions, and 52.2 on CuratedTrec. It also surpassed BART on MS MARCO abstractive QA by +2.6 BLEU and +2.6 ROUGE-L, and obtained 72.5% accuracy on FEVER fact verification. Human evaluation on 452 pairs found RAG more factual than BART in 42.7% of cases and less factual in just 7.1%. Its capacity for “hot-swapping” memory indexes proved that knowledge updates could instantly adjust model facts (e.g., accuracy for world-leader queries rose from 4% to 70% after index update).

Building on this foundation, later surveys and studies [16], [23], [29] explored architectural and optimization advances. The comprehensive survey [23] categorized RAG into naive, advanced, and modular frameworks and reported that specialized models like RAPTOR achieved a 20% accuracy gain on the QuALITY benchmark when paired with GPT-4. It also highlighted multilingual extensions using NLLB-E5, which enabled zero-shot cross-language retrieval via multilingual distillation. The RAG for LLMs Survey [29] found that long-context models which has more than 200,000 tokens now facilitate open-domain reasoning over extended documents and that noise-tolerant retrieval can surprisingly increase accuracy by up to 30% in certain tasks.

The Enhancing RAG Study [29] performed 74 controlled experiments to derive optimization best practices. Its proposed Contrastive In-Context Learning variant (ICL1D+) which achieved FActScore 57.0 on TruthfulQA and 74.4 on MMLU marking a 3.9% ROUGE-L and 2.99% MAUVE improvement over baseline configurations. Conversely, adversarial prompts dropped performance to as low as 7.48 ROUGE-L, emphasizing robustness needs for real-world deployment. Collectively, these studies prove that RAG’s fusion of retrieval and generation enhances factual precision while supporting scalability and multilinguality — qualities essential for Civic.ai as a bilingual, information-grounded citizen assistant.

The RAG framework [8] established a new paradigm for knowledge-intensive NLP tasks. By augmenting generative models with a retrieval component, it ensures outputs remain grounded in factual evidence. Recent surveys [16], [23] systematically categorize RAG variants from dense retrievers like DPR to hybrid retrievers combining dense and sparse indices.

Enhancing RAG best practices [16] explores tuning retrieval granularity, embedding alignment, and document ranking to optimize factuality. These principles are vital for building government-domain chatbots where misinformation risk is critical. Moreover, multilingual RAG implementations are being explored to handle cross-lingual document retrieval [23], a feature that Civic.ai will need for Bangla-English interoperability.

Evaluation of RAG-based systems has also evolved. Benchmarks such as RAGAs [22] automate retrieval-quality scoring, Meanwhile, Ragas Benchmarking on Diploma Theses [26] suggests a real-world evaluation scenario relevant to factual query-answering contexts. Altogether, these studies contribute valuable approaches for assessing accuracy in Civic.ai.

2.2.4 AI in Government and Civic Service Applications

AI-driven government services are spreading globally. The comprehensive review [25] of AI in e-government demonstrates that AI technologies can noticeably improve efficiency, transparency, and citizen engagement in civic services. Citizen-Centric Governance [25] emphasizes participatory governance and AI-enabled transparency.

AI in E-Government [30] identifies challenges such as data standardization, privacy, and organizational readiness, while Trust Frameworks for Government AI [12] emphasizes trust, ethical and accountability specially in use of AI in government services. AI-driven Public Consultation Platforms [27] showcase how local governments integrate AI to improve civic engagement. The focus is on both public servants and developers. They examine specific AI features such as how they are used, what benefits and challenges there are and how public servants view them. Although it’s not a traditional chatbot many features overlap with what a government chatbot would require. The platform includes features like AI powered OCR, content moderation, AI sensemaking , AI translation and human oversight. Meanwhile, the

Discover Artificial Intelligence review [21] summarizes global benefits and risks of AI in government services. It is a systematic literature review. The team searched Scopus and Google Scholar databases for papers relevant to AI in government services, selected around 78 initial papers and after screening they narrowed to 36 that met criteria for detailed analysis. They used organizational theory as a conceptual framework to categorize findings. Additionally, hands-on studies show real world impacts.

Chatbots in Public Service Provision [24] highlights how citizens view and develop trust in digital government agents. Generative AI in Government Chatbots [17] with an advanced experimental approach tests how large language model based civic bots perform to deliver accurate results in civic services.

Mehr [1] demonstrates that AI, more specifically chatbots can make government services faster and more citizen friendly through providing services such as answering questions, processing documents, routing requests, translating information and drafting replies. The study forces that AI should be part of a wider citizen centric strategy and follow clear ethical principles. Together, these studies outline the technical, ethical, and organizational dimensions that inform civic.ai transparency, accessibility, multilingual inclusivity, and policy alignment with data ethics principles.

2.2.5 Evaluation, Benchmarking, and Accuracy

Effective evaluation confirms model reliability and public trust. Foundational NLP benchmarks such as GLUE (General Language Understanding Evaluation) [3] assess general linguistic knowledge across a diverse set of tasks such as question answering, sentiment analysis and textual entailment. Since performance of GLUE quickly overshadowed non-expert human capabilities, a better and more rigorous SuperGLUE [5] benchmark was welcomed. SuperGLUE solves harder tasks and incorporates diverse formats, including complex question answering and coreference resolution and aims to drive innovation by leaving a noticeable performance gap between machines and humans.

For multilingual systems, XQuAD (Cross-lingual Question Answering Dataset) [4] is a crucial benchmark that includes translated SQuAD QA pairs consisting of ten languages. This dataset is vital in terms of testing the model’s ability to maintain deep cross lingual representations necessary for factual accuracy when switching between English and Bangla. TyDi QA [7] evaluates information seeking QA in typologically diverse languages, rather than focusing mostly on English. It provides a high quality benchmark that tests information seeking across multiple languages and encourages development of QA models that generalize across languages. They followed steps such as dataset creation, quality evaluation with analysis and baseline models with experiments.

BERTScore [6] offers semantic similarity scoring using contextual embeddings, addressing limitations of BLEU and ROUGE. The team works to improve evaluation of generated text beyond existing n-gram overlap metrics. According to them the

older metrics fail when there’s lexical variation. They focused on using deep contextual embeddings via BERT to compute similarity more semantically HELM [13] offers holistic evaluation across dimensions such as fairness, robustness, and efficiency. They stress on improving transparency in how language models are evaluated, defining a taxonomy of evaluation scenarios and benchmarking a wide variety of language models under the same standards and settings in order to reveal trade offs and gaps.

MT-Bench and Chatbot Arena [20] evaluate how coherent and engaging a chatbot’s dialogue is by looking at factors like how relevant and fluent the responses are along with user preferences which makes them particularly useful for assessing overall conversational quality from the user’s perspective. Some recent work has also turned attention to domain specific evaluation.

Evaluation Methods for Dialogue Systems [10] offers an intensive survey of subjective and automated metrics. Gan et al. [28] shows a clear and cohesive survey of Retrieval-Augmented Generation (RAG) evaluation methods highlighting accuracy, safety and efficiency. The paper divides evaluation into internal and external introducing metrics such as precision@K, ROUGE and hallucination rate. It reviewed over 30 benchmarks and datasets across domains like law, medicine and finance. It demonstrates how RAG systems can be evaluated for faithfulness, latency and cost.

Es et al. [22] offers RAGAS, a reference free automated evaluation framework for Retrieval-Augmented Generation systems which evaluates faithfulness, answer relevance as well as context relevance using LLM based prompts and embedding similarity without relying on human annotated ground truth. They also present WikiEval, a benchmark dataset for evaluating RAG systems on real world Wikipedia content. RAGAS outperforms baseline methods like GPT Score and GPT Ranking. Collectively, these tools guide Civic.ai’s evaluation plan, ensuring its bilingual and factual dialogue system meets both computational and social accuracy standards.

2.3 Summary of Key Findings

The overall literature collectively shows a converging trajectory in natural language processing and civic technology toward multilingual, retrieval-augmented, and trustworthy conversational systems. How Transformer and LLM-based architectures [2], [9], [14], [19] built the theoretical and technical foundation for context-aware, scalable, and multilingual language modeling, using whom we can develop systems like Civic.ai to understand complex, bilingual citizen queries. Research in conversational AI [11], [15], [18] highlights how end-to-end transformer-based dialogue frameworks can maintain coherent and multi-turn interactions, while innovative verification and cross-checking strategies such as Chain-of-Verification help reduce hallucination which is a crucial issue for factual public guidance. Retrieval-Augmented Generation (RAG) frameworks [8], [16], [23], [29] comes out as a major advancement that connects generative reasoning with grounded retrieval, offering both factual solidity and dynamic updatability. These are the qualities essential for public-facing information systems. Along with these technical advances, studies on AI in governance [1], [12],

[17], [21], [24], [25], [27], [30] reveal that while AI powered platforms can significantly increase transparency, efficiency, and inclusivity, they also come with a price often. As they raise concerns about accountability, ethical governance, and data protection. Lastly, the body of evaluation research [3], [4], [5], [6], [7], [10], [13], [20], [22], [26], [28] gives a comprehensive ecosystem of metrics and benchmarks which range from linguistic accuracy and coherence to factuality and fairness. Basically, they offer tools to assess the performance of systems like Civic.ai across technical and ethical dimensions.

However, the study also reveals a notable research gap. Despite significant advancement in transformer-based architectures [2], [9], [14], [19], conversational AI [11], [15], [18], and retrieval-augmented frameworks [8], [16], [23], [29], most research remains focused on high-resource languages and commercial applications. The existing studies on AI-driven governance [1], [12], [17], [21], [24], [25], [27], [30] prioritize efficiency and transparency but largely overlook inclusivity in multilingual or low-resource settings. While frameworks like RAG have proven effective for factual consistency and real-time information grounding, their integration with bilingual large language models like English and Bangla remains underexplored. Furthermore, evaluation methods [3], [4], [5], [6], [7], [10], [13], [20], [22], [26], [28] offer strong benchmarks for linguistic and factual assessment, yet only a few studies apply these systematically to public-sector AI systems. This creates a critical research gap at the intersection of multilingual LLMs, factual retrieval, and civic engagement, particularly for developing nations such as Bangladesh, where digital governance still faces language accessibility and data scarcity challenges. Therefore, the current study aims to bridge this gap by designing and evaluating Civic.ai, a bilingual, retrieval-augmented AI assistant that supports transparent, accessible, and citizen-centric service interaction within government frameworks.

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